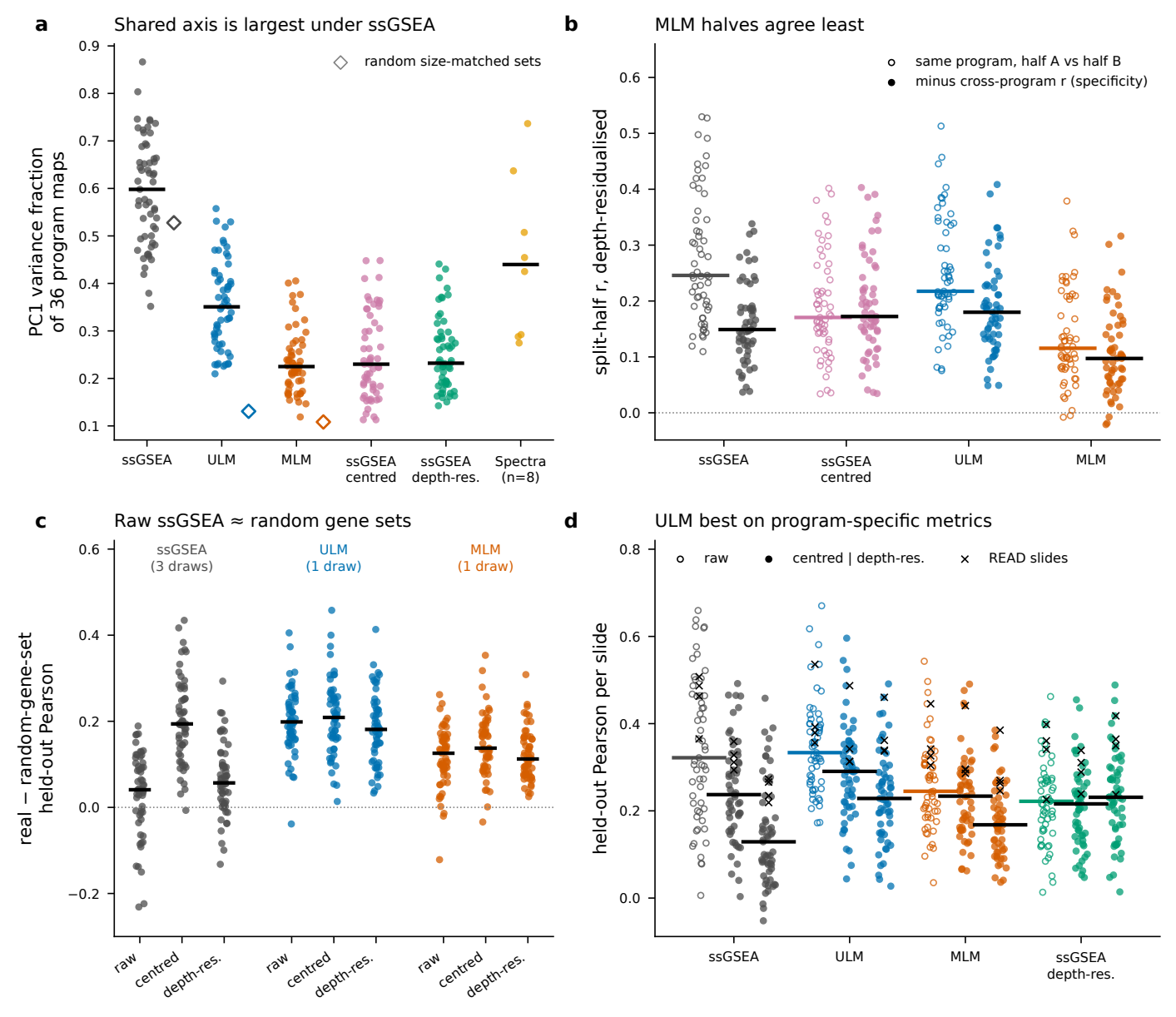

a
Shared axis is largest under ssGSEA
random size-matched sets
PC1 variance fraction of 36 program maps
ssGSEA
ULM
MLM
ssGSEA centred
ssGSEA depth-res.
Spectra (n=8)
b
MLM halves agree least
same program, half A vs half B
minus cross-program r (specificity)
split-half r, depth-residualised
ssGSEA
ssGSEA centred
ULM
MLM
c
Raw ssGSEA ≈ random gene sets
ssGSEA (3 draws)
ULM (1 draw)
MLM (1 draw)
real − random-gene-set held-out Pearson
raw
centred
depth-res.
d
ULM best on program-specific metrics
raw
centred | depth-res.
READ slides
held-out Pearson per slide
ssGSEA
ULM
MLM
ssGSEA depth-res.